

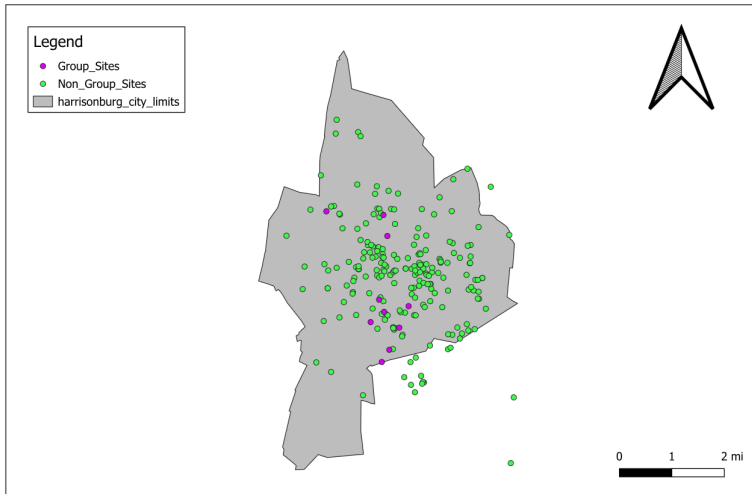
Final Project
A Quantitative Analysis of Air Quality in Harrisonburg

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Data Collection Sites



Our Measurements

Tim's Measurements:

- Taken late at night
- Taken on a humid and stormy day

Sam's Measurements:

- Taken in the early afternoon
- Taken on a sunny and warm day

PM 2.5 Analytics

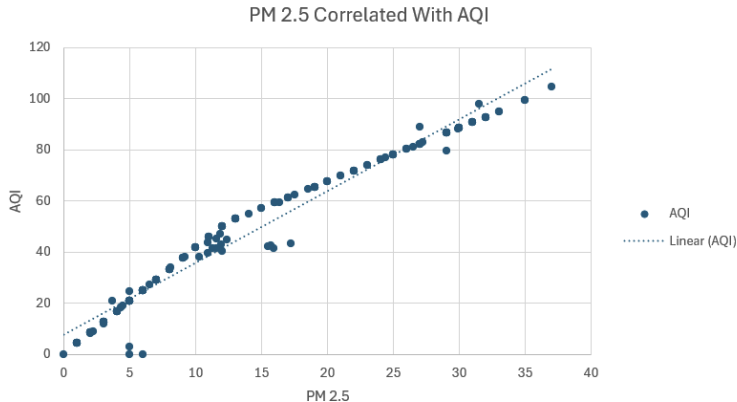
True PM 2.5	Predicted PM 2.5	Absolute Error
5	26.01	21.01
37	19.71	17.29
35	15.55	19.45
32	14.27	17.73
6	14.20	8.20
33	14.19	18.81
7	12.77	5.77
7	12.31	5.31
33	11.69	21.31
5	8.14	3.14

PM 2.5 Analytics

- Median absolute error value of 13.8 micrograms per cubic meter.
- Large differences due to geographic discrepancies in collections.

Other Interesting Results

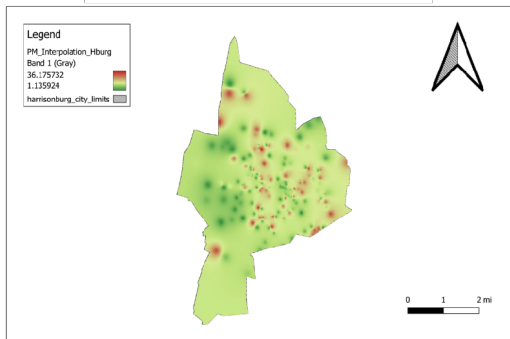
- Correlation between PM 2.5 and AQI: pseudo-logarithmic.



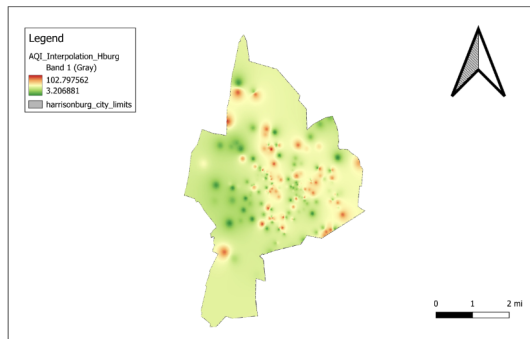
Other Interesting Results

- Correlation between PM 2.5 and AQI: geographically related.

PM 2.5 Interpolated Heatmap

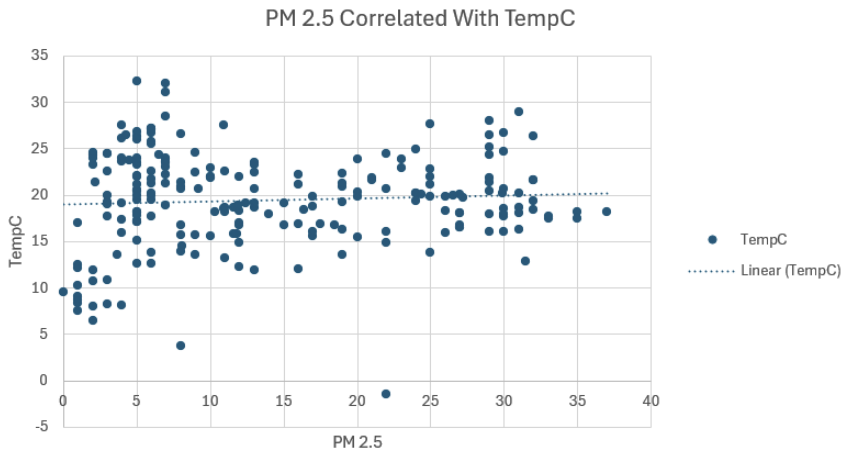


AQI Interpolation Heatmap



Other Interesting Results

- Temperature: completely uncorrelated with PM 2.5.



Final Interesting Results...

Statistic	PM 10	PM 2.5	Ozone	AQI	Temperature (Celsius)
Median	13	11	0.2	42.2	19.8
Maximum	45	37	36.2	104.7	32.2
Minimum	0	0	0	0	-1.5

References

Du, Xu & Varde, Aparna. (2016). Mining PM_{2.5} and traffic conditions for air quality. 33-38. 10.1109/IACS.2016.7476082.

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U.S. Environmental Protection Agency. (2024a). Particulate matter (PM) basics. <https://www.epa.gov/pm-pollution/particulate-matter-pm-basics>

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